Signs of market orders and human dynamics

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Abstract

A time series of signs of market orders has been found to exhibit long memory. There are several proposed explanations for the origin of this phenomenon. A cogent one is that institutions tend to split their large hidden orders into small pieces before execution to prevent the increase in the trading costs. Although, non-professional individual trader with small hidden order may also split her order. Since trading may not be her most pressing concern, the distribution of the interevent time for the execution of her split orders follows bursts and heavy tails. Taking the bursty nature of the human activity patterns into account, we present a mathematical model of order signs that have a long memory property. More precisely, we introduce a discrete time stochastic process for polymer model, and show it's scaled process converges to superposition of a Brownian motion and fractional Brownian motions with Hurst exponents greater than $\frac{1}{2}$.

Keyword: signs of market orders, long memory, burst, fractional Brownian motion, Hurst exponent

Introduction

Empirical studies [2, 4, 6] on high frequency financial data of stock markets that employ the continuous double auction method have revealed a time series of signs of market orders has long memory property. The auto-correlation function of the order signs decays as a power law of the lag and the exponent of the decay is less than 1, which equivalent to a Hurst exponent of the time series is greater than $\frac{1}{2}$. In contrast, a time series of stock returns is known to have short memory property.

There are various explanations for the origin of the long memory property of order signs. Among them, dominant explanation is the trading strategy of large traders who split their hidden orders into small pieces to minimize the trading costs [5].

We have proposed a mathematical model of order signs by adopting large traders' order splitting strategy [3]. Since influence of the trading strategy of small trader is obviously not negligible, we also take their strategies into account in our new model. The distribution of the timing of human activities is found to be characterized by bursts and heavy tails [1]. By expressing a trader's hidden order and timing of executions by a notion of polymer, we construct a discrete time stochastic process. So as to verify analytically our process of order signs has a long memory property, we consider its scale limit. We prove that scaled process converges to superposition of a standard Brownian motion and infinitely many fractional Brownian motions with Hurst exponents greater than $\frac{1}{2}$, that is the limit process has a long memory property. Our tool for obtaining our result is the method of the cluster expansion developed in the mathematical study of statistical mechanics. We use the Kotecký-Preiss theory and the Cauchy formula to prove the convergence of finite dimensional distribution, and use Pfister's argument to prove the tightness.

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